



DEALER BEST PRACTICE SERIES

Bulk Fuel Storage & Handling

Application Maintenance **Component Life Management** **Component Rebuild** Safety **MARC Management**

Bulk Fuel Storage & Handling0

1.0 Introduction1

2.0 Best Practice Description.....1

3.0 Implementation Steps3

4.0 Benefits.....4

5.0 Resources Required4

6.0 Supporting Attachments / References4

7.0 Related Best Practices4

8.0 Acknowledgements.....4

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1.0 Introduction

Mine sites purchase, store, and consume millions of gallons of fuel each year. The quality of fuel is a major factor in machine performance economy and durability. Proper handling and storage can reduce fuel contaminants and contamination related failures. Increasingly complex and sophisticated fuel injection systems require much cleaner fuel than in the past. The use of contaminated fuel in modern fuel systems can cause an unacceptable rate of injector failure and unnecessary repair costs. Proper handling and storage can remove a good portion of fuel contaminants and help improve machine performance.

Since most fuel injector failures or premature wear-outs are the result of contaminated fuel, understanding a few important principles of fuel handling, storage, and tank design will help to reduce unnecessary failures, prolong injector life, maintain fuel economy, and increase machine availability.

2.0 Best Practice Description

The two most common contaminants in the fuel are dirt and water. Excessive dirt causes premature injector wear-out. Excessive water reduces fuel lubricity and can cause injector seizure and contingent engine damage. The two critical factors in fuel quality are:

- The cleanliness of fuel delivered to the site
- The storage and handling practices used after the fuel arrives

Clean fuel delivered to the machine fuel tank is defined by Caterpillar as ISO 18/16/13 or cleaner with a maximum water content of 500 ppm.

In some geographic areas, fuel suppliers to mines have the infrastructure and discipline to deliver relatively clean fuel. However, in the majority of mining applications that is not the case. Fuel leaving the refinery is transported to remote locations through ships, pipelines, barges and trucks and is often stored in intermediate storage tanks. Each time the fuel is stored or moved, it has the opportunity to attract contaminants. In some cases, gross amounts of water and/or dirt can contaminate the fuel. Much of this can be removed through proper storage and handling.

Fuel Storage Capacity

Given adequate storage time, much of the dirt and water in fuel will settle to the bottom of the storage tank if the fuel is left undisturbed. Fuel storage capacity on the site should be adequate to let new fuel deliveries settle for 2-3 days before drawing supply fuel from the tank. New fuel should never be added to the tank while the tank is being used for supply. This only serves to stir up settled contaminants on the bottom of the tank and put them back in suspension, contaminating the supply fuel.

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Bulk Fuel Storage & Handling	DATE 2/17/2015	CHG NO 02	NUMBER 1106-2.10-1037
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Storage Tank Design

Regardless of size or shape, fuel storage tanks should have the following key design features:

- Sloped tank floor
- Well located suction pipe
- Desiccant vent filters
- Non-corrosive tank lining

Sloped Tank Floor

The tank floor should be sloped with a drain at the lowest point. Since water is heavier than fuel, water which settles out of the fuel will flow to the lowest spot in the tank. It can then be easily drained and disposed of. Tanks should be drained frequently to minimize the presence of water on the tank floor. Water provides a breeding ground for bacterial growth, which can cause severe filter plugging on machines and corrosion damage to the storage tank. The frequency of drain intervals depends on ambient temperature and humidity, as well as, the contamination of the fuel being delivered.

Suction Pipe Location

The suction pipe on the tank should be well above floor level in order to ensure that only clean fuel is drawn in, and water and dirt that have settled to the bottom of the tank remain undisturbed. Older design storage tanks often had suction pipes located at or near the bottom of the tank. This kept the tank floor clean by allowing contaminants to flow out with the supply fuel. Most newer design tanks use an elevated suction pipe. Some high quality tanks use a floating suction pipe that rises and falls with the fuel level in the tank. This feature always draws fuel from the highest and cleanest level.

Desiccant Vent Filters

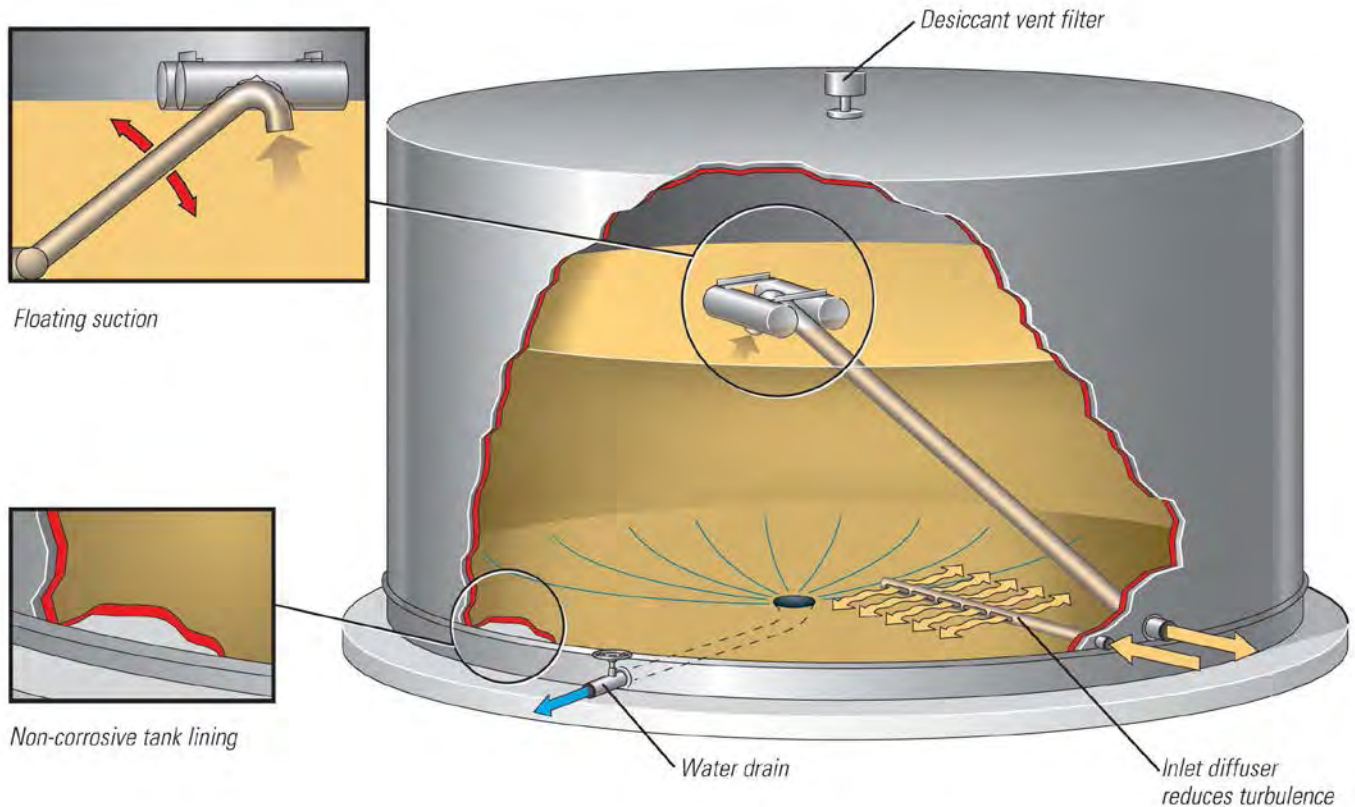
Daily temperature changes along with air drawn into the tank as fuel is withdrawn can cause condensation. Desiccant vent filters should be used on the tank vent in order to remove any moisture from air entering the tank. These filters have a finite capacity and must be replaced when the desiccant is exhausted.

Tank Lining

Tanks should be epoxy coated on the inside to prevent corrosion. At a minimum, the tank floor and the bottom three feet of the tank sides should be epoxy coated to resist corrosion from trapped water.

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Bulk Fuel Storage & Handling	DATE 2/17/2015	CHG NO 02	NUMBER 1106-2.10-1037
------------------------------	-------------------	-----------------	--------------------------



3.0 Implementation Steps

Major fuel suppliers have the technical expertise to provide guidance on tank design and sizing. The most common problem on most mine sites is lack of tank capacity. The mine capacity and haulage fleet often grows faster than the fuel storage facility. This usually results in fuel being added to tanks while it is being consumed. This practice stirs up contaminants and keeps them in suspension in the delivery fuel. There are three key steps to proper fuel storage and handling.

- Adequate capacity to allow fuel to settle
- Proper tank design
- Fuel handling maintenance

Fuel Handling Maintenance

It is important to have an adequate number of trained personnel responsible for making sure fuel is handled properly. This includes:

- Using particle counters and water detection tests to measure fuel cleanliness.
- Sampling incoming shipments
- Assuring proper settling time and draining tanks to remove water
- Maintaining desiccant filters and any supplemental bulk fuel filtration equipment
- Sampling outgoing fuel going into machine tanks.
- Assuring fuel quality in remote day tanks and mobile fuel trucks

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Bulk Fuel Storage & Handling	DATE 2/17/2015	CHG NO 02	NUMBER 1106-2.10-1037
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4.0 Benefits

Maintaining ISO 18/16/13 or better fuel cleanliness with a maximum of 1,000 ppm water should allow most 3500 Series injectors to last to engine overhaul.

- If 18/16/13 cannot be achieved with proper tank capacity and fuel handling, the use of bulk fuel filtration is recommended to achieve needed fuel cleanliness. Bulk filtration equipment is available through Caterpillar.

5.0 Resources Required

Cost information can be obtained from the tank constructor prior to construction of new tanks.

6.0 Supporting Attachments / References

For more information on bulk fuel filtration equipment, refer to Caterpillar publication:

Caterpillar Bulk Fuel Filtration Equipment

PEHJ0156

For more information on the effects of water and dirt on fuel injector life, bulk fuel filtration, and fuel storage and handling, refer to the following publications (available in print form only):

Improving Component Durability – Fuel Systems

SENR9620

Improving Component Durability – Managing Fluid Cleanliness

SEBF1020

7.0 Related Best Practices

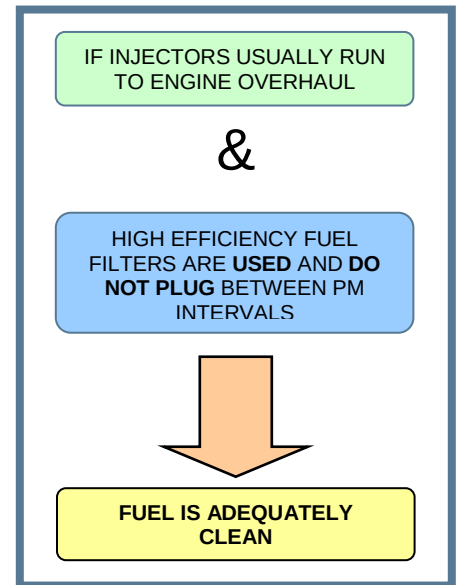
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Fluid Cleanliness Management

8.0 Acknowledgements

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